Muhammad Ahmad Amin

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Education

BS Computer Science | National University of Sciences and Technology - NUST (2024 - 28)

Skills

Programming Languages | Java, Python, JavaScript
Data Science | Python, NumPy, Pandas, Matplotlib, Seaborn
Frontend Development | HTML, CSS, JavaScript, React.js, TailwindCSS
Backend Development | Node.js, Express.js, REST APIs, Postman (tool)
Databases | MySQL (Relational Database/SQL), MongoDB (noSQL), Mongoose
Version Control System | Git, GitHub

Experience

AI Research Assistant | NUST SEECS — July 2026 - Present

- Conducting research on **efficient deep learning models** for real-time image classification on resource-constrained mobile devices, targeting agricultural applications in Pakistan
- Evaluating lightweight **CNN architectures** for deployment on low-end Android hardware under real-world field conditions

Projects

FactArena | Full-Stack AI-Judged 1v1 Debate Platform (Argument Settler) — Live Demo | GitHub

- Built a **competitive AI debate** platform where two players argue a topic in turns and an **impartial AI judge** analyzes the full transcript to declare a winner with individual scores and reasoning.
- Engineered a **turn-based** argument system with a synchronized countdown timer that automatically submits the debate transcript to an **LLM judge** on expiry.
- Built with React, Node.js, Express, and integrated Groq API for impartial judging. **Deployed** on Netlify and Render.

Linracy | Full Stack MERN Social Media Platform — Live Demo | GitHub

- Built a **full-stack** social media platform with user **JWT-based authentication**, profiles, photo uploads via Cloudinary and **post management** using React.js, Node.js, Express and MongoDB **(MERN Stack)**.
- Built a sleek, **responsive** user interface optimized for both desktop and mobile using modern frontend design practices. **Deployed** frontend on Netlify and backend on Render.

O3Scope | Ozone Analysis, Visualization and Prediction Using Nasa Dataset — Hex App | GitHub

- Built an **interactive Hex app** visualizing global and city-level ozone trends (2012–2026) with risk zones.
- Developed linear regression **predictions for 2027 ozone levels** using Hex AI-assisted analysis.
- Processed **NASA OMPS HDF5 datasets** and created **dynamic visualizations** with Matplotlib & Cartopy.

Open Source Contributions

hnn-core | Human Neocortical Neurosolver — GitHub

Merged PR: Improved GUI usability by refactoring button implementation to standard Widget for consistency.

Problem Solving & Competitive Programming — Leetcode Profile

Solved 415+ Data Structures and Algorithms Problems on Leetcode and ranked in the top 3% globally among 15M+ users.

Hackathons

Hex-a-thon | Built O3Scope, an interactive ozone data analysis, visualization and prediction tool using NASA datasets